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FY-MCA Division: A

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1. Write a program to read a file and count how many vowels are present in its content.

   Solution:

```
Class_Codes > Assignment_fileOperations > 01_file_countVowels.py > ...
1   # 1. Write a program to read a file and count how many vowels are present in its content.
2   name = input("Enter file name: ")
3   with open(name, "r") as f:
4     data = f.read()
5     vowels = "aeiouAEIOU"
6     count = 0
7     for char in vowels:
8       count += data.count(char)
9     print("Vowels present in the file: ", count)
10
```

   Output:

```
Problems  Output  Debug Console  Terminal  Ports                                   powershell
PS D:\Projects\MCA\Semester_01\Python\Class_Codes\Assignment_fileOperations> python .\01_file_countVowels.py
Enter file name: random.txt
Vowels present in the file:  30
PS D:\Projects\MCA\Semester_01\Python\Class_Codes\Assignment_fileOperations>
```

2. Read all lines from a file into a list. Print the total number of lines and display the 2nd and 4th line.

   Solution:

```
Class_Codes > Assignment_fileOperations > 02_file_totalLines.py > ...
1   # 2. Read all lines from a file into a list. Print the total number of lines and display the 2nd
    and 4th line
2
3   #name = "random.txt"            #if using "name" as 1st parameter in open()
4   # or: name = input("Enter file name: ")         # ""
5
6   with open("random.txt", "r") as f:
7       lines = f.readlines()
8       print("Total number of lines in the file:", len(lines))
9       if len(lines) >= 4:
10          print("The 2nd line is:", lines[1].strip())
11          print("The 4th line is:", lines[3].strip())
12      else:
13          print("The file does not have enough lines to display the 2nd and 4th lines.")
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports                                    powershell
PS D:\Projects\MCA\Semester_01\Python\Class_Codes\Assignment_fileOperations> python .\02_file_totalLines.py
Total number of lines in the file: 5
The 2nd line is: This is the second line.
The 4th line is: This is line 4.
PS D:\Projects\MCA\Semester_01\Python\Class_Codes\Assignment_fileOperations>
```

3. Take a sentence from the user, split it into words, and store them in a list.

   Print the longest word.

   Solution:

```
Class_Codes > Assignment_fileOperations > 03_file_printLongestWord.py > ...
1   # 3. Take a sentence from the user, split it into words, and store them in a list. Print the
    longest word.
2
3   sentence = input("Enter a sentence: ")
4   words = sentence.split()
5   longest_word = max(words, key=len)
6   print("The longest word in the sentence is:", longest_word)
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports                                    powershell
PS D:\Projects\MCA\Semester_01\Python\Class_Codes\Assignment_fileOperations> python .\03_file_printLongestWord.py
Enter a sentence: Hello world, Kalsubai was my most favourite trek ever!
The longest word in the sentence is: favourite
PS D:\Projects\MCA\Semester_01\Python\Class_Codes\Assignment_fileOperations>
```

4. Read a file, split its content into words, and store them in a list.
   i. Print the total number of words.
   ii. Print all unique words.

Solution:

```
Class_Codes > Assignment_fileOperations > 04_file_splitContentIntoWords.py > ...
1   # 4. Read a file, split its content into words, and store them in a list.
2   #   i. Print the total number of words
3   #   ii. Print all unique words
4
5   file_name = input("Enter the file name: ")
6   with open(file_name, "r") as f:
7       content = f.read()
8       words = content.split()
9       print("Total number of words in the file:", len(words))
10      unique_words = set(words)
11      print("Unique words in the file:", unique_words)
```

Output:

```
Problems  Output  Debug Console  Terminal  Ports                                    powershell

PS D:\Projects\MCA\Semester_01\Python\Class_Codes\Assignment_fileOperations> python .\04_file_splitContentIntoWords.py
Enter the file name: random.txt
Total number of words in the file: 18
Unique words in the file: {'line', ':)', 'file', 'line.', 'operations', 'in', 'Byeee', 'This', 'World!', 'second', 'Hello',
 'is', 'Executing', 'the', 'python.', '4.'}
PS D:\Projects\MCA\Semester_01\Python\Class_Codes\Assignment_fileOperations>
```